Data Dictionary

Column	Definition	Type
`police_force_name`	Name of the police force that recorded the crime. One of four: West Midlands, Staffordshire, Nottinghamshire or Essex.	`object`
`month`	Calendar month the crime was recorded, stored as the first day of that month.	`datetime64`
`crime_type`	Crime category as published by police.uk, for example Burglary or Shoplifting.	`category`
`crime_count`	Number of crimes recorded for that force, month and category.	`int64`
`year`	Calendar year taken from the month field. Supports filtering and the population join.	`int32`
`month_number`	Calendar month number from 1 to 12, taken from the month field. Lines up the same month across different years for seasonal comparison.	`int32`
`total_population`	Resident population of the force area for that year, from the ONS mid year estimates.	`int64`
`crime_rate_per_1000`	Crimes per 1,000 residents, calculated as crime_count divided by total_population, multiplied by 1,000.	`float64`
`mean_imd_decile`	Average deprivation decile of the LSOAs within the force area. Runs from 1 to 10 where 1 is the most deprived. A lower value therefore means a more deprived force.	`float64`

General

Grain. One row is a unique combination of police force, month and crime type. Each row therefore reports the number of crimes of one category, recorded by one force, in one month.

The dataset covers four police forces across 35 months and holds 1,960 rows. Uniqueness at this level is enforced by a validation check in the pipeline.

Sources

- UK Police crime data: https://data.police.uk/data/
- ONS population estimates for police force areas: https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/adhocs/3194populationestimatesforpoliceforceareasinenglandandwalesbysingleyearofageandsexmid1991tomid2024
- English Indices of Deprivation 2025, File 7: https://www.gov.uk/government/statistics/english-indices-of-deprivation-2025
- ONS Local Authority District to Police Force Area lookup: https://ckan.publishing.service.gov.uk/dataset/local-authority-district-to-community-safety-partnership-to-pfa-december-2021-lookup-in-ew

The lookup is not an analytical source. It is used only to map each LSOA to a police force so that deprivation can be summarised at force level.

Assumptions

- One row in the source crime files represents one recorded crime, so counts are row counts.
- The day component of the month field carries no meaning. It is a convention of the date type, as the source gives the year and month only.
- Population estimates are annual and end at mid 2024, so 2025 and 2026 use the 2024 figure. This affects recent rates by under 2 per cent.
- Deprivation is treated as fixed across the period, since the 2025 indices are a single snapshot.
- Population is held unrounded so the rate uses full precision. ONS ask for rounding to the nearest 100 when a population figure is quoted directly, so this is left to the reporting layer.

Extensive detail on every calculation, assumption and validation check is documented in the pipeline notebook or the .py file.